Supplementary Diagnostics: Do Optimized Personas Predict New Respondents?

Anonymous Author(s)
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CCS Concepts

• **Computing methodologies** → **Intelligent agents**.

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1 Source-data identity audit

A separate CPU audit compared the derived corpus with the original IPIP120.dat from Johnson's data repository. The article describes the IPIP-NEO-120 study [2]; the later OSF deposit linked here supplies the audited source file. The original file contains 619,150 records. All 410,168 retained derived records matched exactly by case identifier and the 120 recoded item values: zero absent cases and zero item mismatches. The source file's SHA-256 was checked against its published OSF metadata, and the derived corpus's SHA-256 matched the frozen experiment input before and after the audit. Only aggregate counts were exported; respondent records used private temporary storage. The original DAT120 codebook confirms that reverse-keyed items were already recoded and zero denotes missingness. No second reversal was applied. This verifies retained case/item identity, not demographic-field equivalence, completeness of the historical filtering procedure, participant consent, upstream data permissions, or absence from pretraining. The audit did not estimate predictive performance or inspect panel-specific outcomes.

2 Scope

The evidence map in Table S3 distinguishes the analysis families. These additional analyses are post hoc, using the exposed original 600-person cohort and three search seeds. They do not supply a new confirmatory sample. Original 110-test results are unchanged. The initial representation tests use one 33-test Holm family; direct objective and interaction tests use a separate nine-test family. The figure/table differences are computed from unrounded values. Intervals for LLM comparisons cross respondents and observed search seeds; sign-flip p-values condition on those seeds. Ridge-only contrasts resample respondents with trained predictors held fixed.


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Objective contrasts. The first four contrasts are raw-fitness minus calibrated-fitness RPS. Positive values favor calibrated fitness. Remaining rows subtract these paired objective effects; they are not inferred from subgroup significance. All five interaction intervals contain zero. No effect is claimed to generalize to an unseen optimizer seed. The smallest possible two-sided sign-test p-value with three seed pairs is .25; the table instead uses respondent-level sign flips after averaging the three seeds.

3 Representation controls

Categorical ridge uses 175 one-hot features: five categories for each of 30 facets and five traits. Continuous ridge uses the same 35 values before discretization, standardized on training people. Both use 8,000 training and 120 selection respondents, choosing penalty from $\{1, 10, 100, 1000\}$ before reading evaluation labels. Selected penalties are 100 and 1, respectively. Categorical, continuous and raw-history ridge score 0.11384, 0.11558 and 0.10955. Their pairwise differences (categorical–raw, continuous–raw, categorical–continuous) are 0.00428, 0.00603 and −0.00174; corresponding 95% intervals are $[.00378, .00478]$, $[.00540, .00664]$ and $[-.00231, -.00120]$. All have Holm $p = .006599$. These are fitted-model gaps, not a causal decomposition or an information bound.

4 Search ranking diagnostic

Each observed pool contributes the first complete evaluation per distinct candidate. We compare the saved raw and cross-fitted-calibrated scores, excluding tied pairs from the discordance denominator. The 24 pools span both models, three seeds, two optimizers and two objectives. Median Spearman correlation is .344, median discordant fraction .356, and median overlap of the two top-five sets is zero. These are within-search descriptors, not independent hypothesis tests or evidence of causal mediation.

Independent summary recomputation. A separate implementation reconstructed the 48 mixture-screen and 24 prior-anchored extension comparisons from the retained individual loss arrays. For all 72 contrasts it exactly reproduced the mean differences, 5,000-draw respondent-bootstrap intervals, conditional sign-flip probabilities, Holm adjustments and joint screen decisions. The same declared random stream was used for numerical comparison. This check validates aggregation from saved losses; it does not regenerate model forecasts, refit predictors, or provide independent confirmation of the adaptive scientific findings. The aggregate receipt and checking source accompany the evidence bundle; individual losses remain on the research host.

5 Artifact inventory

Machine-readable aggregates contain all 33 representation contrasts and all 9 objective contrasts, plus 24 history summaries. The

Table S1: Raw and calibrated losses of the same selected templates; the final column lists three seed-specific calibrated differences from each cell's base.

Model	Search family, fitness	Raw RPS	Calibrated RPS	Differences by seed
Qwen3.6	GA, raw	0.15665	0.13127	+0.00132, +0.00134, +0.00087
Qwen3.6	GA, calibrated	0.16377	0.12965	-0.00066, +0.00003, -0.00069
Qwen3.6	MAP-Elites, raw	0.15577	0.13132	-0.00053, +0.00045, +0.00378
Qwen3.6	MAP-Elites, calibrated	0.17042	0.12976	-0.00029, -0.00087, +0.00015
Granite	GA, raw	0.17209	0.13520	-0.00090, -0.00195, -0.00123
Granite	GA, calibrated	0.18219	0.13391	-0.00295, -0.00280, -0.00223
Granite	MAP-Elites, raw	0.17640	0.13594	-0.00202, -0.00096, +0.00110
Granite	MAP-Elites, calibrated	0.18436	0.13389	-0.00203, -0.00385, -0.00214

Table S2: All nine post-hoc objective contrasts. Intervals cross respondents and observed seeds; Holm p values condition on the three frozen runs.

Contrast	Difference	Crossed 95% interval	Conditional p_{H}
Qwen:GA:raw–calibrated	+0.00163	[+0.00111,+0.00219]	0.002
Qwen:MAP:raw–calibrated	+0.00156	[-0.00012,+0.00347]	0.002
Granite:GA:raw–calibrated	+0.00130	[+0.00060,+0.00204]	0.002
Granite:MAP:raw–calibrated	+0.00205	[+0.00015,+0.00340]	0.002
Qwen: GA–MAP objective	+0.00007	[-0.00190,+0.00206]	0.719
Granite: GA–MAP objective	-0.00075	[-0.00249,+0.00185]	0.002
GA:Qwen–Granite	+0.00033	[-0.00028,+0.00099]	0.252
MAP:Qwen–Granite	-0.00049	[-0.00158,+0.00049]	0.047
Optimizer–model interaction	+0.00082	[-0.00029,+0.00211]	0.012

fixed diagnostic protocol, analysis scripts and figure scripts accompany this local draft. Raw human records remain on the research host. Their inclusion in an anonymous release requires a separate permission and de-identification review; this local inventory is not a claim that a public release already exists.

Table S3: Distinct analysis families. Original post-hoc families reuse the exposed Evaluate600; fixed-template transfer uses 600 additional people from the same corpus. Holm corrections are within families, not across the adaptive research program.

Question / family	Status	Target and interpretation
Primary and original arm contrasts / 110	Frozen before evaluation	Respondent mean over three frozen runs; crossed intervals also resample observed runs.
Representation / 33	Post hoc	Fitted-model comparison at 8,000 labels, same exposed people.
Objectives and interactions / 9	Post hoc	Direct run-index paired effects, conditional sign flips; three seeds.
Ordinary budget / 48; shared-prior budget / 48	Sequential adaptive diagnostics	Fixed fitted predictors at each budget, three LLM runs; separate families.
Mixture screen / 48	Adaptive exploratory	Respondent effects conditional on nine fixed fit/run combinations; no optimizer or subset resampling.
Anchored rival / 24	Subsequent adaptive extension	Frozen original hybrid versus prior-anchored ridge and its neutral mixture; same nine fixed combinations.
Readout replay / 4 intervals	Engineering diagnostic	Calibration-only numerical compatibility, separate from predictive improvement.
Selected-template replay / 4 primary + 4 secondary	Frozen replay on exposed data	Two fresh-server rounds of the GA objective contrast; separate Holm-four families. No new optimizer runs or people.
Additional-respondent transfer / 3 primary	Fixed after original outcomes	Historical GA/MAP winners on 600 inventoried-disjoint same-corpus people; three Granite tests, conditional on old seeds. Qwen and ridge are mandatory diagnostics.
Same-shortlist selector / 10	Subsequent exploratory intervention	Five candidates from each of 12 archived GA pools; raw versus cross-fitted calibrated selection, then the old exposed Evaluate600. Four selector effects, four search effects, two interactions; three old seeds.

6 Full method visualization

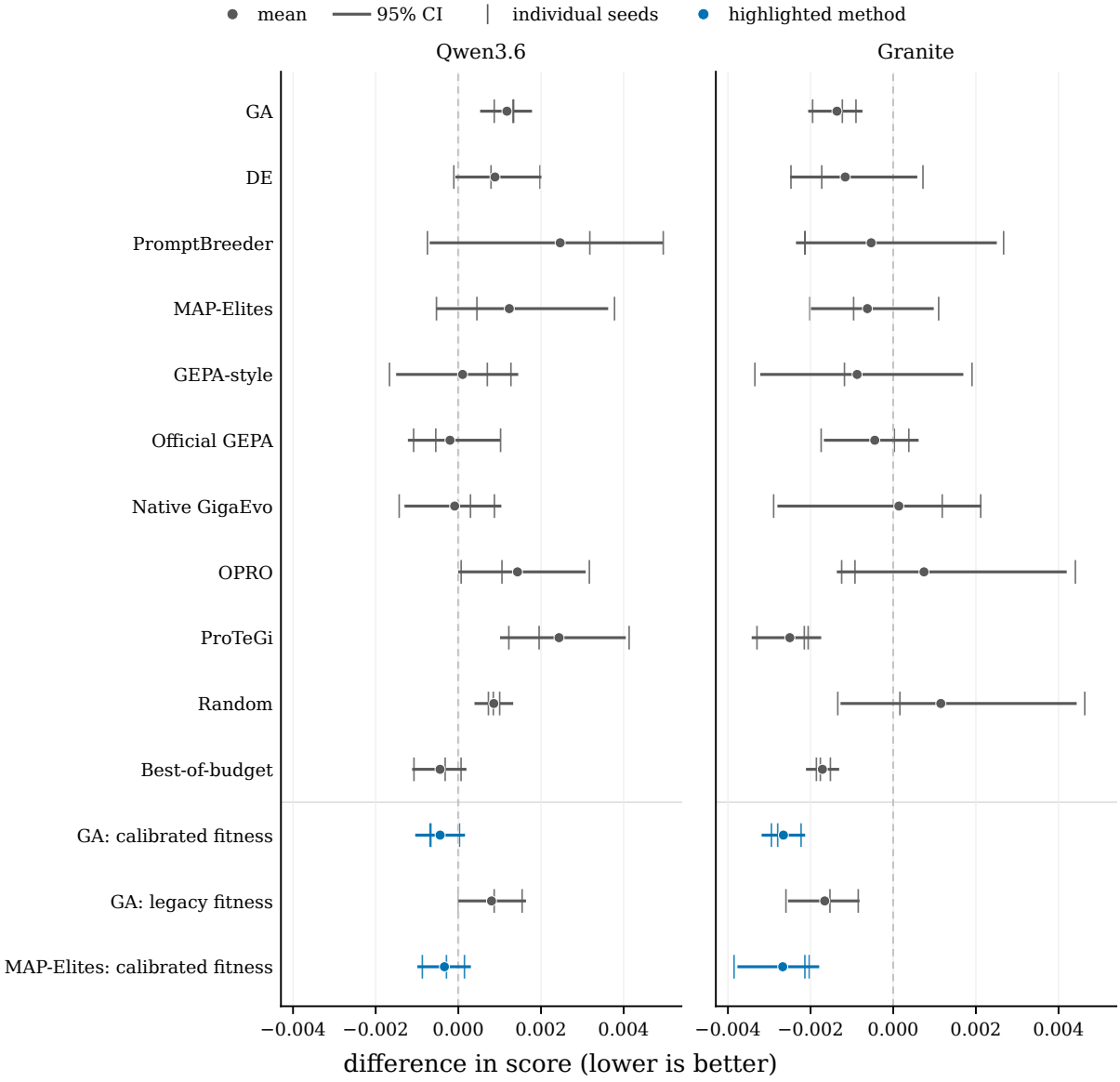


Figure S1. All 14 method/objective identities relative to their own base. Circles are mean differences, bars are crossed 95% intervals, and ticks are seed values. Blue highlights calibrated-fitness variants. The family remains the original 110 comparisons.

7 Paired MAP-Elites objective effects

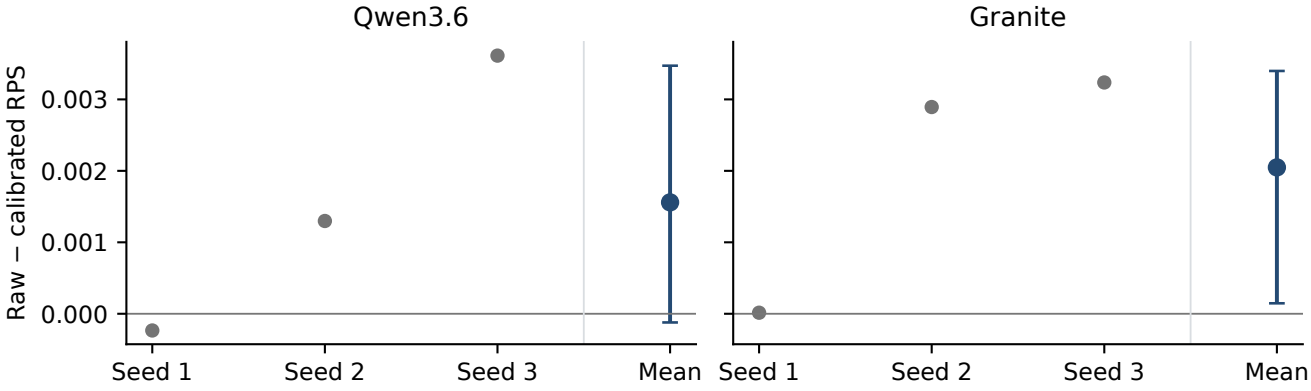


Figure S2. All paired MAP-Elites seed effects and overall crossed interval. Seeds are paired by initialization index, not a shared mutator-sample trajectory. Points show independent search runs. Positive favors calibrated fitness.

8 Calibration-only readout repeatability

An engineering replay used the same neutral persona, pinned source and serving configuration, and all 120 calibration people twice per model. No search or new evaluation-person forecast was run. A pointwise probability tolerance of 5×10^{-4} failed; archived original runs also fail that tolerance against their shared reference. Before reading the new full-panel replay results, a dated amendment fixed a separate calibration-score compatibility criterion: four paired respondent-bootstrap 98.75% intervals, each wholly within $[-0.0001, +0.0001]$ RPS (20,000 draws; Bonferroni-four interval levels). The margin is below 8% of the smaller exploratory GA objective effect. This is a post-failure engineering amendment, not an original scientific preregistration or a claim of population equivalence.

Source, model, configuration, current token reconstruction, repeated token sequences, person/item order, probabilities and costs passed their checks. Three raw-RPS comparisons passed the amended interval rule. Qwen repeat 2 had mean difference +0.0000700 and interval $[-0.0001570, +0.0002709]$, which failed. All four comparisons and failed pointwise probes were retained; the planned new-seed search was not released. The observed score variation does not establish its cause, invalidate every original effect estimate, or bound optimized-candidate variability. Historical token sequences were not saved, so present reconstruction cannot retrospectively prove their identity.

Table S4: Calibration replay versus historical neutral readout. Intervals are 98.75% paired respondent intervals; the fixed compatibility margin is ±0.0001. These differences are not optimized-template uncertainty bounds.

Replay	Mean raw RPS change	Interval	Within margin
Qwen 1	+0.0000121	[-0.0000012,+0.0000364]	Yes
Qwen 2	+0.0000700	[-0.0001570,+0.0002709]	No
Granite 1	+0.0000011	[0,+0.0000034]	Yes
Granite 2	+0.0000001	[-0.0000036,+0.0000030]	Yes

For scale, the saved three-run GA raw-minus-calibrated effects are 0.00163 (Qwen) and 0.00130 (Granite); the planned Granite MAP-Elites effect is 0.00205. The replay intervals concern neutral calibration forecasts, whereas these scientific effects concern selected-template evaluation forecasts. The saved forecast scores reproduce exactly; serving-level repeatability remains open.

Concurrency intervention. A subsequent frozen engineering protocol restarted the Qwen server for four blocks with request concurrency 4, 1, 1, 4, each replaying the same 120-person neutral calibration panel twice. All 57,600 forecasts completed with identical prompt-token hashes. Within-server respondent RPS root-mean-square differences were 0.000118, 0.000979, 0.001071, and 0.000084, respectively. Serial requests therefore did not repair repeatability in this diagnostic. Six of eight historical-reference intervals extended beyond the original ±0.0001 margin, including both replays of the final concurrency-four block despite its small within-server difference. These intervals are descriptive diagnostics, not a new multiplicity-controlled acceptance test. Two blocks per setting do not identify a particular numerical kernel as the cause or bound optimized-template variation. The additional-seed scientific campaign remains blocked; no failed replay was replaced or tolerance relaxed.

9 Selected-template serving robustness

A separate protocol fixed two complete replay rounds before new inference. Each round covers both models, both GA fitness objectives and all three historical template seeds: 24 cells in total, each with a fresh server and the unchanged final runner. Selected templates, model locks,

input panels, source hashes, the 8,000-person prior and historical results were preserved. Each cell repeats base/evolved calibration on 120 people and base/evolved/wrong-profile prediction on the same exposed 600 people. There is no new search, prospective respondent sample or hybrid-mixture test.

The primary analysis refits calibration on the original calibration panel. The secondary analysis transfers each historical calibration weight and CDF center to the new evaluation forecasts. Each regime has four model-by-round contrasts and a separate Holm-four correction. The frozen positive screen requires all four positive differences, positive lower crossed-interval bounds and adjusted $p < .05$. Intervals use 5,000 crossed respondent/template draws; sign flips condition on the same three templates. Neither two serving rounds nor resampling those templates estimates variation across new optimization runs.

Table S5: Complete frozen-template serving replay. Positive raw-minus-calibrated GA RPS favors calibrated fitness. Crossed 95% intervals resample respondents and the three fixed templates; respondent sign-flip p values are Holm-adjusted separately across four comparisons per calibration regime. These are two serving rounds on the same exposed people, not new optimizer runs.

Calibration	Model	Round	Difference	95% interval	Holm p
Refitted	Qwen	1	+0.00161	[+0.00109,+0.00215]	0.0008
Refitted	Qwen	2	+0.00159	[+0.00108,+0.00212]	0.0008
Refitted	Granite	1	+0.00130	[+0.00059,+0.00203]	0.0008
Refitted	Granite	2	+0.00130	[+0.00059,+0.00203]	0.0008
Fixed	Qwen	1	+0.00162	[+0.00109,+0.00215]	0.0008
Fixed	Qwen	2	+0.00160	[+0.00109,+0.00213]	0.0008
Fixed	Granite	1	+0.00130	[+0.00059,+0.00203]	0.0008
Fixed	Granite	2	+0.00130	[+0.00059,+0.00203]	0.0008

Both regimes satisfy their four-contrast screen. All 12 seed-specific primary differences are positive; the complete seed values and cell-level controls are retained in the aggregate report. Calibration weights do not change in any of the 48 base/evolved comparisons with historical values, although CDF centers and probabilities can change. Across the 24 base/evolved evaluations per model, mean refitted-RPS changes versus history range from −0.0000493 to +0.0000783 for Qwen and −0.00000281 to +0.00000109 for Granite. Corresponding raw-RPS changes range from −0.0000851 to +0.0001075 and from −0.00000352 to +0.00000581, respectively. The largest individual probability-coordinate drift is 0.4337 for Qwen and 0.0617 for Granite; preserved aggregate contrasts therefore do not establish pointwise reproduction. Wrong-profile minus evolved RPS remains positive in every cell (Qwen 0.04078–0.04810; Granite 0.03164–0.03782). Ridge fits and their saved losses are unchanged by this experiment.

All 24 cells complete with 2,937,600 confirmed forecasts, zero failed or uncertain forecasts, and 3,596,132,160 reported prompt tokens. Slurm records 29,223 allocated single-GPU seconds (8.1175 GPU-hours), including server startup. This is replay cost, not an end-to-end optimization-efficiency estimate. All four primary contrasts were additionally recomputed from saved individual losses by an independent statistics implementation; its receipt reports the numerical agreement. The historical failed engineering gates remain failed. These results support the observed GA contrast under the two tested serving replays, not a repaired serving configuration or independent confirmation on new people, optimizer trajectories, or behavioral outcomes.

10 Additional-respondent fixed-template transfer

After the original outcomes were inspected, a separate internal protocol froze 24 historical selected GA/MAP-Elites-style templates: both models, two search fitness objectives and three old seeds. The new 600 complete-case respondents were sampled from 36,119 eligible people after exclusion of the reachable inventoried historical exposure union. They come from the same public IPIP120 corpus, so lifetime or model-pretraining exposure cannot be excluded. Original Train8000 priors, Calibrate120 alpha and CDF centers, frozen prompts, model snapshots and the selected same-history ridge specification were reused. No new search or shortlist selection occurred; this test does not evaluate the separate developmental aligned-selector recipe. The internal freeze is not an external preregistration.

The three declared primary tests are Granite calibrated MAP versus raw MAP, calibrated MAP versus its neutral base, and calibrated GA versus its neutral base. Positive differences below are comparator minus calibrated-template RPS. Crossed intervals resample respondents and the three *observed* search seeds; sign flips condition on those seeds. The original combined decision rule requires a positive crossed interval and Holm $p < .05$ within the three-test family. Qwen and other comparisons are diagnostics with their uncorrected sign-flip p values shown for completeness; their small values are not additional declared primary successes.

Table S6: All eight E1 objective and neutral comparisons on the 600 additional respondents. Positive favors the calibrated historical winner. Primary Holm p applies only to the three Granite tests marked P; the Granite MAP calibrated-versus-raw interval crosses zero and fails the combined rule.

Model	Arm	Contrast	Δ RPS [crossed 95% CI]	Sign-flip p	Primary p_H
Qwen	GA	cal vs raw	+0.001805 [+0.001358, +0.002257]	.0002	–
Qwen	GA	cal vs base	+0.000337 [−0.000174, +0.000854]	.1184	–
Qwen	MAP	cal vs raw	+0.001680 [+0.000007, +0.003393]	.0002	–
Qwen	MAP	cal vs base	+0.000180 [−0.000506, +0.000806]	.4043	–
Granite	GA	cal vs raw	+0.001229 [+0.000626, +0.001812]	.0002	–
Granite	GA	cal vs base (P)	+0.002286 [+0.001661, +0.002893]	.0002	.0006
Granite	MAP	cal vs raw (P)	+0.001523 [−0.000182, +0.002733]	.0002	.0006
Granite	MAP	cal vs base (P)	+0.002156 [+0.001344, +0.003113]	.0002	.0006

Table S7: Mandatory E1 controls for each calibrated historical winner. Values are comparator minus calibrated-template RPS with crossed 95% intervals; negative ridge differences mean ridge is better. These diagnostics were not in the three-test primary family.

Model	Arm	Same-history ridge minus template	Wrong-persona minus matched template
Qwen	GA	−0.020850 [−0.021898, −0.019877]	+0.044778 [+0.041452, +0.048465]
Qwen	MAP	−0.020995 [−0.022161, −0.019898]	+0.041762 [+0.039263, +0.044285]
Granite	GA	−0.025188 [−0.026441, −0.023977]	+0.035653 [+0.032601, +0.038636]
Granite	MAP	−0.025318 [−0.026688, −0.023941]	+0.035213 [+0.031855, +0.038961]

All 24 cells have 600 people and 60 target items. The independent output gate confirmed the original frozen lineage, 24 immutable attempts, serving logs, probability and mass arrays, and 2,937,600 requested and confirmed forecasts with no failed or uncertain request. Prompt-token receipts sum to 3,500,088,480; these are new inference costs, not search costs. The separately written finite CPU verifier reconstructed all 72 labelled cell/control score rows, the selected ridge, all 16 contrast distributions and the three-test Holm correction from saved raw forecasts and target labels; its maximum discrepancy from the frozen analysis was 8.33×10^{-17}. It shares the pinned CDF projection and ridge-fit numerical primitives and does not independently regenerate model forecasts.

The same-history ridge, selected originally and refit on 8,000 old training people with the same 60 observed answers, scores 0.10821 RPS here. Calibrated GA/MAP means are 0.12906/0.12920 on Qwen and 0.13340/0.13353 on Granite. The three observed Granite MAP seed effects against raw are −0.000288, +0.002381, +0.002477; the first is unfavorable. A low answer-token mass diagnostic for raw Granite MAP seed 3 is retained (mean 0.9473; 2.37% of target cells below 0.5), above the frozen mean-mass exclusion threshold. Neither favorable base comparisons nor matched-versus-wrong-persona signals establish optimizer superiority to ridge, new-search generalization, or behavioral transfer.

11 Exploratory same-shortlist selector intervention

This E2 experiment reused the *original exposed* Evaluate600, rather than E1's additional 600 people. For each of the 12 archived GA trajectories (two models, raw/calibrated search objectives, three seeds), the same five historical candidates received new Select120 forecasts. Raw RPS and a fixed two-fold cross-fitted calibrated RPS each selected one candidate from that identical pool. Both choices were frozen before fresh Calibrate120/Evaluate600 inference, with a fresh neutral-base readback; one coincident pair was deduplicated. There was no new search or new respondent cohort. The selection stage confirmed 432,000 forecasts and the final stage 1,512,000, with zero failed or uncertain requests; prompt-token receipts were 624,369,600 and 1,843,379,400, respectively. These are additional serving costs, not total optimizer costs.

In Table S8, selector effects subtract raw-selector RPS from calibrated-selector RPS within the named search objective. Search effects subtract raw-search RPS from calibrated-search RPS within the named selector. Negative values favor the first, calibrated option. The interaction is the calibrated-selector search effect minus the raw-selector search effect. Crossed 95% intervals resample respondents and the three *observed* search seeds; respondent sign flips condition on those seeds, with Holm over all ten two-sided tests. A favorable claim requires both a crossed interval excluding zero and adjusted $p < .05$. The three seed columns are differences on the fixed, paired archived trajectories, not new-search replicates.

Table S8: All ten E2 contrasts on the exposed original Evaluate600. Values are calibrated RPS differences as defined in text; lower is better. All four direct selector intervals include zero. The Qwen search effect under the raw selector is the only interval wholly below zero; it is not selector benefit.

Model	Contrast	Δ [crossed 95% CI]	Seed 1	Seed 2	Seed 3	p_H
Qwen	Selector; calibrated search	−0.000153 [−0.000495, +0.000181]	+0.000044	−0.000434	−0.000068	.3179
Qwen	Selector; raw search	−0.000587 [−0.001456, +0.000044]	+0.000000	−0.001541	−0.000221	.0020
Qwen	Search; calibrated selector	−0.000740 [−0.001712, +0.000308]	−0.001699	+0.000376	−0.000896	.0020
Qwen	Search; raw selector	−0.001174 [−0.001852, −0.000562]	−0.001743	−0.000730	−0.001049	.0020
Qwen	Search × selector	+0.000434 [−0.000152, +0.001131]	+0.000044	+0.001107	+0.000153	.0144
Granite	Selector; calibrated search	−0.000013 [−0.000208, +0.000302]	+0.000249	−0.000134	−0.000154	.8352
Granite	Selector; raw search	−0.000473 [−0.001334, +0.000104]	+0.000062	−0.001435	−0.000045	.0020
Granite	Search; calibrated selector	−0.000171 [−0.001693, +0.001081]	−0.001782	+0.001124	+0.000143	.3179
Granite	Search; raw selector	−0.000631 [−0.001872, +0.000357]	−0.001969	−0.000177	+0.000252	.0020
Granite	Search × selector	+0.000460 [−0.000158, +0.001250]	+0.000187	+0.001301	−0.000109	.0020

Calibrated selection changed 11 of 12 winners but none of its four direct effects passes the combined interval-and-p rule. The two raw-search selector tests have small adjusted sign-flip probabilities despite crossed intervals spanning zero; they are not declared improvements. Original selection-audit attempt 12149 failed on a post-receipt server shutdown diagnostic. Versioned selection gate 13521 classified that event explicitly and passed the archived arrays; final gate 13527 and independent CPU readback 13549 passed all 12 cells and ten contrasts. This operational chain does not repair numerical serving compatibility. Compared with the historical base, fresh Qwen base per-cell mean absolute RPS-loss drift was roughly 0.00031–0.00037 (maximum individual drift 0.0028–0.0048); Granite drift was much smaller. All reported E2 contrasts therefore compare predictions within the fresh serving execution. E2 does not test its selected winners on E1's additional people, and E1 never evaluated those winners.

12 Third-panel Granite-80 frozen-mixture transfer

This E3 test was designed after inspection of the original, E1, and E2 outcomes. Its internal protocol fixed the old Granite calibrated-GA n=80 recipe, controls, four contrasts, 0.001-RPS margin and inference before opening the third panel's target answers; it is not an external preregistration. The 600 complete-case respondents were sampled from 35,519 eligible cases after exclusion of 374,649 reachable inventoried case IDs, including every original split and E1's additional Eval600. They are new to these project panels, not a new sample recruited outside the public corpus; unknown lifetime and model-pretraining exposure remain possible. The old selected prompts, neutral prompts, Granite model snapshot, first-Cal60 scale and CDF centers, old 60-person mixture-selection weights, penalties, categorical/raw features, three disjoint 80-person ridge-fitting subsets, and 8,000-person population prior were reused unchanged. Ridge penalties were chosen on old Select120. The persona search consumed Search80 and Select120 targets, and each new respondent supplies 60 observed answers; "80-person" names only the paired ridge-fit budget. The third-panel labels selected none of these choices.

For each representation the optimized persona plus ordinary ridge was compared with neutral persona plus prior-anchored ridge and neutral persona plus the *same* ordinary ridge. Each neutral arm retained its own old selected mixture weight. The first control changes prompt, weight and ridge centering; the second changes prompt and weight while holding ridge coefficients fixed. Neither is an isolated prompt-only causal contrast. The four planned comparator-minus-hybrid RPS gains form one two-sided Holm family. Before paired respondent resampling, nine fixed per-person loss vectors (three old search seeds by three fixed subset seeds) were averaged, without ensembling forecasts or treating the nine combinations as independent respondents. The fixed success rule requires each gain to be at least 0.001, its paired respondent-bootstrap lower 95% bound to exceed zero, and its Holm-adjusted respondent sign-flip $p < .05$. Both controls must pass for a representation; uncertainty conditions on those old choices. The three n=80 subset seeds have pairwise overlap 0/0/0 and union 240 within Train8000; the same sets are reused across representations and LLM seeds. This index fact does not turn the nine fitted combinations into independent respondent panels or new search replications.

Table S9: All four declared E3 contrasts on the third 600-person panel. Positive gains favor the old optimized mixture; all pass the fixed gate. The two-sided Holm family contains exactly these four rows.

Features	Neutral comparator	Gain	Paired 95% CI	p_H
Categorical	Prior-anchored	.001465	[.001213,.001722]	.0008
Categorical	Same ordinary	.001843	[.001595,.002088]	.0008
Raw	Prior-anchored	.001251	[.001016,.001490]	.0008
Raw	Same ordinary	.001747	[.001519,.001974]	.0008

Table S10: E3 mean RPS component and mixture controls, lower is better. Each row averages the nine fixed per-person loss vectors. The common full 8,000-person same-history ridge scores 0.10995, better than both hybrids.

Predictor	Categorical	Raw
Optimized mixture	.132355	.131532
Neutral + prior-anchored ridge	.133821	.132784
Neutral + same ordinary ridge	.134198	.133279
Optimized persona alone	.134180	.134180
Neutral persona alone	.136691	.136691
Prior-anchored ridge alone	.138871	.137463
Ordinary 80-person ridge alone	.140831	.139406
Full 8,000-person ridge	.109952	.109952

Table S11: Every E3 old-search-seed/subset combination. Subset indices 1–3 denote seeds 202610081–083. w_O, w_N, w_A are the frozen optimized, same-ordinary neutral and anchored-neutral persona weights; H, N, A are their respective third-panel mean RPS values. The nine rows per representation are descriptive fixed combinations, not independent replications.

Features	Search/subset	w_O	w_N	w_A	H	N	A
Categorical	1/1	.60	.55	.55	.13229	.13421	.13373
Categorical	1/2	.65	.60	.50	.13227	.13415	.13389
Categorical	1/3	.60	.55	.50	.13228	.13423	.13384
Categorical	2/1	.65	.55	.55	.13235	.13421	.13373
Categorical	2/2	.70	.60	.50	.13236	.13415	.13389
Categorical	2/3	.65	.55	.50	.13239	.13423	.13384
Categorical	3/1	.60	.55	.55	.13239	.13421	.13373
Categorical	3/2	.65	.60	.50	.13240	.13415	.13389
Categorical	3/3	.60	.55	.50	.13246	.13423	.13384
Raw	1/1	.60	.50	.50	.13158	.13356	.13295
Raw	1/2	.60	.50	.45	.13137	.13318	.13275
Raw	1/3	.60	.50	.45	.13125	.13310	.13265
Raw	2/1	.60	.50	.50	.13178	.13356	.13295
Raw	2/2	.60	.50	.45	.13156	.13318	.13275
Raw	2/3	.60	.50	.45	.13146	.13310	.13265
Raw	3/1	.55	.50	.50	.13179	.13356	.13295
Raw	3/2	.55	.50	.45	.13154	.13318	.13275
Raw	3/3	.55	.50	.45	.13146	.13310	.13265

The frozen E3 preparation receipt has SHA256 `76f8a7b57f8b867936978244761056073930981c37d750a0b60b4673b9a6bf32`; the new panel's canonical hash is `415583f53de73b7f0295f2c8a289cad1bab311d9f0ce033a21ba840d6e1f8671`. Preparation reproduced the old recipe and old-Eval losses before serving. The independent output gate passed three immutable one-GPU attempts, each with 72,000 requested/confirmed item forecasts and zero failed or uncertain requests. The total was 216,000 forecasts and 237,089,520 reported prompt tokens, excluding historical search, old calibration and CPU fitting cost. The three allocated single-GPU elapsed times were 694, 622 and 640 seconds (0.5433 GPU-hours total). Seed 3 recorded a post-receipt shutdown diagnostic; its forecast arrays and counts passed the versioned gate, so this is not silently removed. The independently written finite-CPU numerical verifier recomputed all four gains, intervals, Holm values, component means and the full-ridge control from archived raw forecasts and labels. Its maximum absolute discrepancy was 9.38×10^{-11}, within the protocol's fixed 10^{-10} tolerance. This readback did not regenerate model forecasts. Neither this operational pass nor the favorable contrasts validate new search trajectories, the E2 selector winners, Qwen serving compatibility, external behavioral transfer, or superiority to the full ridge.

13 Post hoc fixed-weight E3 readback

E4 reuses the already exposed E3 Eval600, frozen Granite prompts, saved probabilities and n=80 ordinary ridge coefficients. Its question and analysis were fixed internally only *after* E3 effects had been inspected; no new respondent panel or model inference was used. At the primary fixed weight $w = .5$, each arm keeps its own historical first-Cal60 scale and CDF center, but evolved and neutral forecasts share exactly the same ordinary ridge and numeric weight. This removes unequal mixture weights from the comparison. It identifies the effect of substituting an *arm-specific calibrated forecast package*, not prompt wording alone. The old last-Cal60 weight-selection labels are part of E3's lineage but their selected weights are unused by this fixed-half primary comparison.

For each representation we average nine fixed *per-person losses* (three historical search seeds by three fixed ridge subsets) before pairing the 600 respondents. The two neutral-minus-evolved fixed-half effects form a separate, two-sided Holm family. The internally fixed descriptive gate requires gain at least 0.001 RPS, positive paired respondent-bootstrap lower 95% bound and adjusted respondent sign-flip $p < .05$. With 5,000 draws and seed 20261021, both pass (Table S12). Intervals condition on these old seeds/subsets and the exposed panel. The numerical verifier recomputed every saved loss and statistic from the raw E3 forecasts with maximum discrepancy zero; it shares the frozen CDF/RPS primitives and does not regenerate model forecasts or create prospective confirmation.

Table S12: E4 fixed-half calibrated-package comparisons on exposed E3 respondents ($N = 600$). Positive is neutral minus evolved RPS, lower scores being better. Both rows form the Holm-2 family and pass the descriptive 0.001-RPS/positive-CI/adjusted-p rule.

Features	Evolved	Neutral	Gain [paired 95% CI]	p_{H}
Categorical	.132912	.134389	.001477 [.001291,.001676]	.0004
Raw	.131831	.133279	.001448 [.001264,.001643]	.0004

Table S13: Complete *fixed*, unselected E4 weight grid. Each entry is the mean of nine fixed per-person loss vectors. Weight zero reproduces the same ordinary ridge exactly for both arms. Other weights are descriptive; no best weight was chosen from E3 labels.

Features	w	Evolved RPS	Neutral RPS	Neutral − evolved
Categorical	0	.140831	.140831	.000000
Categorical	.25	.135723	.136517	.000794
Categorical	.50	.132912	.134389	.001477
Categorical	.75	.132398	.134447	.002049
Categorical	1	.134180	.136691	.002512
Raw	0	.139406	.139406	.000000
Raw	.25	.134378	.135150	.000772
Raw	.50	.131831	.133279	.001448
Raw	.75	.131765	.133793	.002028
Raw	1	.134180	.136691	.002512

Table S14: Descriptive fixed-half sensitivity to the three *old* Granite search seeds. "Leave out" averages the remaining two seed means, each already averaged across the three fixed n=80 ridge subsets. All values are neutral-minus-evolved RPS; these are not new optimizer replications.

Features	Seed 1	Seed 2	Seed 3	Leave out 1	Leave out 2	Leave out 3
Categorical	.001579	.001317	.001535	.001426	.001557	.001448
Raw	.001546	.001295	.001504	.001399	.001525	.001420

At the same $w = .5$ but with raw *uncalibrated* persona CDFs, the descriptive neutral-minus-evolved gain is .005634 categorical (evolved .141848, neutral .147482) and .005579 raw (evolved .140851, neutral .146430). These gains are larger than the primary calibrated-package differences, yet both arms' absolute RPS values are worse than their calibrated half-weight counterparts. This secondary comparison has no additional claim threshold or corrected-test status. It neither proves that calibration caused a particular prompt effect nor isolates prompt text from other arm-specific forecast differences.

The three E3 ridge subsets are reconstructed from the exact old 80-of-8,000 seeded permutations; pairwise intersections are 0/0/0 and their union is 240 people. They sit *inside* Train8000, are reused for both representations and all three prompt seeds, and add no unique labelled people beyond Train8000. Before evaluation, the original disjoint Train8000, Search80, Select120 and Calibrate120 roles comprise 8,320 people and 499,200 target-item labels (8,320 × 60). Search80 supplied prompt fitness, Select120 shortlist and ridge-penalty selection, Train8000 supplied the population target prior and full-ridge reference, and Calibrate120 was split 60/60 for E3 calibration and historical weights. Each E3 respondent then supplied 60 observed input answers and 60 separate evaluation targets. E1 and E3 each added 600 inventoried-disjoint same-corpus evaluation people; E4 adds none. These role and item counts are not an equal-supervision comparison of the persona and its n=80 ridge component.

The selected three historical Granite calibrated-GA cells used by E3 account for 1,627,200 successful forecasts, 2,158,592,520 reported prompt tokens and 9.5253 allocated GPU-hours. They are *included in*, rather than added to, the 84-cell original campaign total in Table S15. E3 separately added 216,000 forecasts, 237,089,520 prompt tokens and 0.5433 allocated GPU-hours; E4 added zero GPU forecasts or prompt

tokens and used finite CPU jobs. Its freeze, analysis and verifier allocated 96 CPU-seconds in total; the separate historical cost join allocated 8 CPU-seconds. These are allocated core-time, not measured CPU utilization. The independent E4 receipt binds analysis SHA256 `b2d6b3950a6899ba19d535046179eef84dd2471f1fa36b4fef9808bbf61feb71`, the saved individual-loss archive, both primary effects, the fixed grid, and the inherited E3 costs.

Table S15: Source-bound original 84-cell historical cost. Forecast and prompt counts cover successful receipts; GPU time counts task allocations, including failed original search tasks. The five linked smoke/probe jobs have measured allocation but incomplete inference/token accounting. The selected three Granite cells are already inside the 84-cell rows.

Scope	Confirmed forecasts	Prompt tokens	GPU-h
Search	32,237,760	43,017,723,760	215.856
Selection	3,024,000	4,158,424,800	included
Scoped repair allocation	included	included	54.839
Final	10,281,600	11,979,632,160	28.729
84-cell array total	45,543,360	59,155,780,720	299.424
Linked smoke/probe jobs	unavailable	unavailable	1.176
Measured historical minimum	at least 45,543,360	at least 59,155,780,720	300.599

The 84-cell join required all 84 original search, 12 repair and 84 final Slurm task rows. Search had 72 completed, two failed and ten cancelled task rows; repair and final completed. GPU-hours sum allocated GPUs times elapsed seconds, including startup, and exclude queue time and duplicate batch rows. Earlier smoke jobs 8692 and 8808 and context probe 8918 contribute another 4,233 GPU-seconds, linked by contemporaneous queue/recovery records. This 300.599-hour minimum predates and is outside the current-cycle 60 GPU-hour planning cap. Other historical replays, failed/cancelled request work, mutator/completion tokens, CPU work and dollars are unresolved. Successful receipt totals are not all incurred serving work, and no end-to-end efficiency ranking follows.

14 Unexecuted E5 replication and old-only engineering

A planned replication with three new Granite optimizer seeds and a disjoint 600-person same-corpus panel (E5) remains unexecuted: its strict readout gate failed before search release. No E5 optimizer search, new-panel forecast or holdout-target analysis was run. The engineering checks in Table S16 use the original Cal120 forecasts and runtime metadata only; they add neither a transfer sample nor a new optimizer result. The reference remains the original calibrated-GA Granite seed-1 `base_cal` archive (SHA-256 prefix `9ebeffc8f789`), with the unchanged rule $\max |\Delta p| \leq 5 \times 10^{-4}$.

Table S16: Old-only E5 engineering and independent readback. Completion or verification of a diagnostic does not pass historical compatibility or release search. Profile comparisons use their common respondent rows.

Job	Recorded result
23135	Two fresh sequential servers, explicit batch-invariant OFF/ON: 30,720 confirmed forecasts and 24,258,640 prompt tokens. OFF fails five of six profile pairs; ON probabilities are byte-identical in all six. Both arms fail all four comparisons with the canonical reference; the maximum probability difference remains 0.0303456.
23637	Independent CPU readback of archived raw top-20 log probabilities and reconstruction of answer mass and normalized arrays, with checks of prompt tokens, source lineage and costs, agrees with the diagnostic; historical compatibility still fails.
23645	All 42 historical Granite neutral-Cal arrays, two preserved replays and eight OFF/ON profiles are retained. The canonical exact mode recurs in six historical cells. The 42 historical full-Cal arrays have 27 exact modes but one shared exact tail over respondents 9–120.
23654	Fixed initialization-to-post-capture overlap across all 42 launches: 0/4 overlapping pairs and 27/857 nonoverlapping pairs share a mode; proportion difference -0.0315053, two-sided task-label permutation $p = 1$ (100,000 draws). The fixed overlap proxy has no positive support.

The invariant ON path stabilizes this execution without recovering the historical reference; it does not identify a single numerical kernel. Repeated canonical bytes weaken a one-off corruption hypothesis without proving absence or cause. The overlap test is observational, has only four exposed pairs, and assumes task-label exchangeability that temporal or launch-order dependence may violate. Its interval definition was not changed after the result. Recorded selected attention backend, AOT action and graph mode are constant across the 42 launches despite the multiple output modes. An older fresh-server full-Cal first call and repeat with identical rendered tokens (job 9149) already both

fail the same canonical gate; this is not a new E5 replication. These checks establish no new-search-seed robustness or mechanism-level cause. Full receipt hashes and claim locators are retained in `E5_ENGINEERING_PAPER_BINDING_20261024.json`.

The additional E5 engineering allocations (jobs 23036, 23121 and 23135) consume 0.188333 GPU-hours, increasing the prior current-cycle total from 13.796111 to 13.984444 GPU-hours; job 23135 accounts for 0.094444 GPU-hours of that increment. The independent CPU checks add no GPU allocation. A later mechanism-trace allocation (24116) fails after one second, before server launch or forecasts, adding 0.000278 GPU-hours and bringing the current-cycle total to 13.984722 GPU-hours. Its client-environment preflight validation requests Triton package metadata unavailable in that interpreter; no hidden-state, logit or sampler observation is obtained. That attempt is closed without a retry and supplies no layer-localization evidence. The separately inventoried historical minimum remains 300.599444 GPU-hours (Table S15); it is outside this current-cycle total, and the older job 9149 is outside the current-cycle ledger. Neither total is an end-to-end project cost or an efficiency result.

A subsequent raw-only CPU check (24086) tests the narrower hypothesis of unchanged digit-relative values plus a shared row offset and final round-to-nearest FP32 storage. In the OFF full-Cal repeat contrast, 8 of 28,800 anchored digit components across four forecasts and two respondents exceed the stated component-specific rounding bounds; the maximum change is 0.250000119, versus a maximum bound of 1.91×10^{-6}. All six within-ON contrasts are exactly zero. This rejects that stated offset-and-final-rounding model, not every possible internal arithmetic error, and identifies neither a kernel nor a canonical repair. The failed JSON publication of its earlier implementation (24083) is retained separately and is not interpreted as a numerical result.

Static opcode inspection, without deserializing or executing the AOT payload, finds 21 embedded source texts in the current cached artifact at the historically recorded path; all match the collected current sources (15 vLLM and six Torch/Transformers files). This is a partial cached-source witness for the compiled forward and support code. No October 7 artifact hash exists, so it does not authenticate the historical bytes or the complete scheduler, attention-kernel and runtime environment. Engineering-only preflight failures (24108, 24110, 24112) and the successful CPU compile check (24113) are retained; none is scientific parity evidence.

A separately authorized corrected trace (24315) validates Triton in the pinned vLLM interpreter and the supervisor in its client interpreter (CPU preflight 24313). One default-OFF server performs exactly two old-Cal120 reads: 14,400 confirmed forecasts and 11,369,760 prompt tokens. The GPU job remains FAILED because its final join expects external request IDs, while vLLM adds an eight-hex-character internal suffix. Its 14,400 captured rows and raw artifacts are retained unchanged. A source-derived, posthoc CPU-only binding correction (24328) requires unique raw and normalized IDs, an exact 14,400-row client/server bijection, recomputed request hashes, and unchanged artifact/source hashes before and after analysis. The two additional source files explaining the internal suffix are current post-run witnesses, absent from the original seven-source runtime receipt. The earlier CPU binding implementation failure (24322, before array analysis) is retained.

The bound trace contains 7,200 paired forecasts. Ten pairs differ already in the exact BF16 LM-head-input hidden-state bits; all ten also differ in returned top-logprob records. This localizes observed repeat variation at or before that hidden-state boundary under instrumentation, without identifying an earlier layer or kernel. All ten have different final sampler-call shapes and repeat-local call ordinals, but these contexts also differ widely among unchanged pairs and do not identify a causal batch effect. The other 7,190 pairs have identical hidden states and five captured canonical-digit logits; their 71,900 anchored digit-pair comparisons show no sampler-repeat variation beyond the stated bound. No isolated head variation is observed with identical hidden states. These nulls cover only selected coordinates and the stated arithmetic model. Snapshot kernels and retained buffers can alter scheduling; this trace does not establish the uninstrumented historical cause, repair canonical compatibility, or release E5 search. The one GPU allocation lasts 134 seconds (0.037222 GPU-hours), bringing the current-cycle total to 14.021944 GPU-hours. No additional GPU replay or new-target inference is performed for the CPU binding correction.

Retained-artifact CPU forensics (24574), prespecified over all 7,200 pairs, finds zero token-signature, token-count or selected-position mismatches; every selected position equals the prompt token count minus one. These are receipt checks: the actual token-ID arrays were not retained. All ten changed hidden-state pairs lie in the previously fixed first-eight-respondent group (10/480 forecasts), versus 0/6,720 in the other 112 respondents. Each changed vector differs numerically in 3,390–3,763 of 4,096 coordinates, with no signed-zero-only changes; the least-squares common-scale residual is 0.7595–1.4154% of the second vector's norm. These descriptive metrics do not identify RMSNorm, KV-cache or batch scheduling as a cause. The tensors correspond to selected final-RMSNorm outputs supplied as LM-head inputs; the snapshots are queued only after the head call returns, and backend nonmutation is not independently proved. Earlier-layer states were not retained, so the earliest upstream divergence remains unobserved.

A CPU-only static AOT feasibility audit (24710; 203 seconds, two CPUs, 8 GiB, no GPU) inspects the unchanged 17,075,197-byte cached object using pickle opcodes and Python ASTs without deserializing or executing payloads. It inventories seven complete and five incomplete pickle units, 185 AST-readable Python-source records (34 proof-eligible), and 63 opaque binary units, including a 162,874-byte execution-code witness. The bounded audit does not prove complete producer–buffer–consumer maps for the 66 boundaries, complete original numeric-launch association, or an exact existing logits-index delivery channel; the runner's dynamic input expansion remains unresolved. The verdict is FEASIBILITY_HOLD / OBSERVER_UNAVAILABLE, not a proof that another static method could never recover these relationships. The fixed protocol therefore stops before observer implementation or GPU submission, with no eager fallback, signature change or numerical recompile.

15 Full-data representation comparison

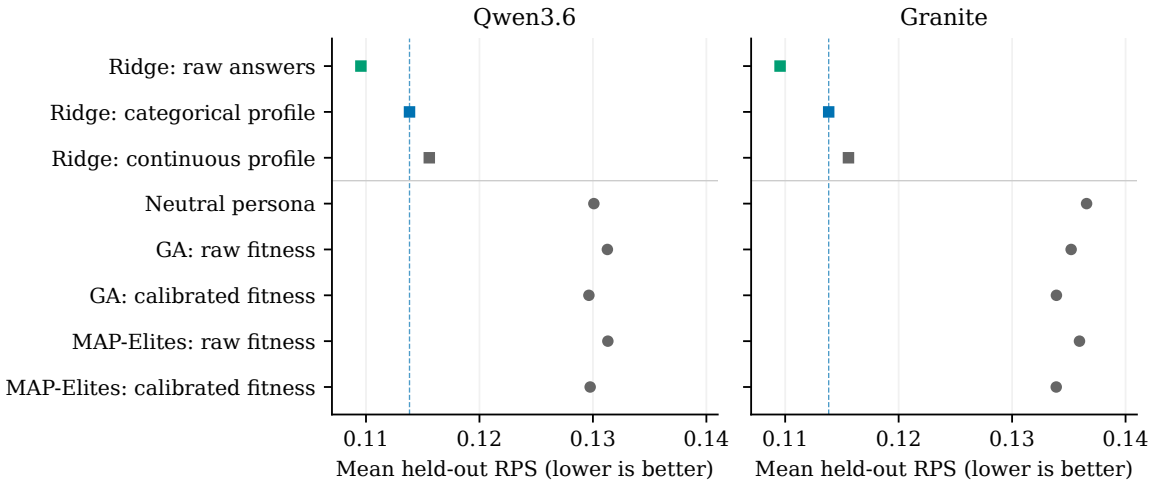


Figure S3. Full-data ridge and persona controls. Shared statistical predictors are repeated across panels for comparison. All ridge controls in this figure use 8,000 person-level labelled mappings. LLM arms use their frozen population calibration; ridge directly predicts ordinal CDFs. These are held-out RPS values, not an equal-supervision comparison. The dashed vertical line marks categorical-profile ridge.

16 Associative label-budget diagnostics

Two additional 48-test families cover the ordinary and shared-prior budget diagnostics, separately from the earlier 33 representation and nine objective contrasts.

The original 600-person cohort was already exposed. The ordinary-ridge diagnostic was followed by a separate rival test using the 8,000-person empirical item prior. Both are post hoc. For each fitting subset, the anchored variant predicts

$$\widehat{F}_{\text{anchored}}(x) = \text{Proj}\big(\widehat{F}_{\text{ridge}}(x) + F_{0,8000} - \overline{Y}_{\text{subset}}\big), \tag{1}$$

where $\widehat{F}_{\text{ridge}}$ is the unprojected intercept-fit ridge output and $\overline{Y}_{\text{subset}}$ the mean target-threshold vector. This changes only the intercept at a given penalty; each variant selects its penalty on the same 120 selection people. It is algebraically centered no-intercept ridge about the population prior. Using the original solver precision makes the full 8,000-point correction zero and exactly reproduces every original individual loss.

Table S17: Mean RPS at each ridge fitting budget. Lower is better.

Features	Fit people	Ordinary	Shared prior
categorical	80	0.13956	0.13802
categorical	200	0.13150	0.13099
categorical	320	0.12791	0.12759
categorical	1,000	0.12069	0.12059
categorical	8,000	0.11384	0.11384
categorical	exact 80	0.13814	0.13669
raw	80	0.13795	0.13643
raw	200	0.12972	0.12921
raw	320	0.12607	0.12575
raw	1,000	0.11824	0.11815
raw	8,000	0.10955	0.10955
raw	exact 80	0.13624	0.13494

Positive contrast values in the accompanying diagnostic files mean LLM minus ridge RPS; negative values favour the LLM. Every variant has 48 comparisons: 2 representations times 6 fitting conditions times 4 GA model/objective arms. Each family is separately Holm corrected. This does not correct for adaptively choosing the second diagnostic after seeing the first. The full 48-row tables, all subset means and frozen selection specifications are retained in the local research artifact; no public data-release claim is made.

The ten fitting subsets per budget overlap. Their mean losses average separate predictors, not ensemble probabilities; their min/max spread is not a confidence interval. Crossed intervals resample only respondents and LLM search seeds, conditioning on statistical fits

and the population prior. Fitting counts omit 120 selection people. LLMs additionally use 120 calibration people and foundation-model pretraining, so matching the empirical prior does not imply identical total supervision or prove a causal sample-efficiency advantage. The exact 80 row uses the original search people instead of training-panel subsets.

17 Fitting-budget visualization

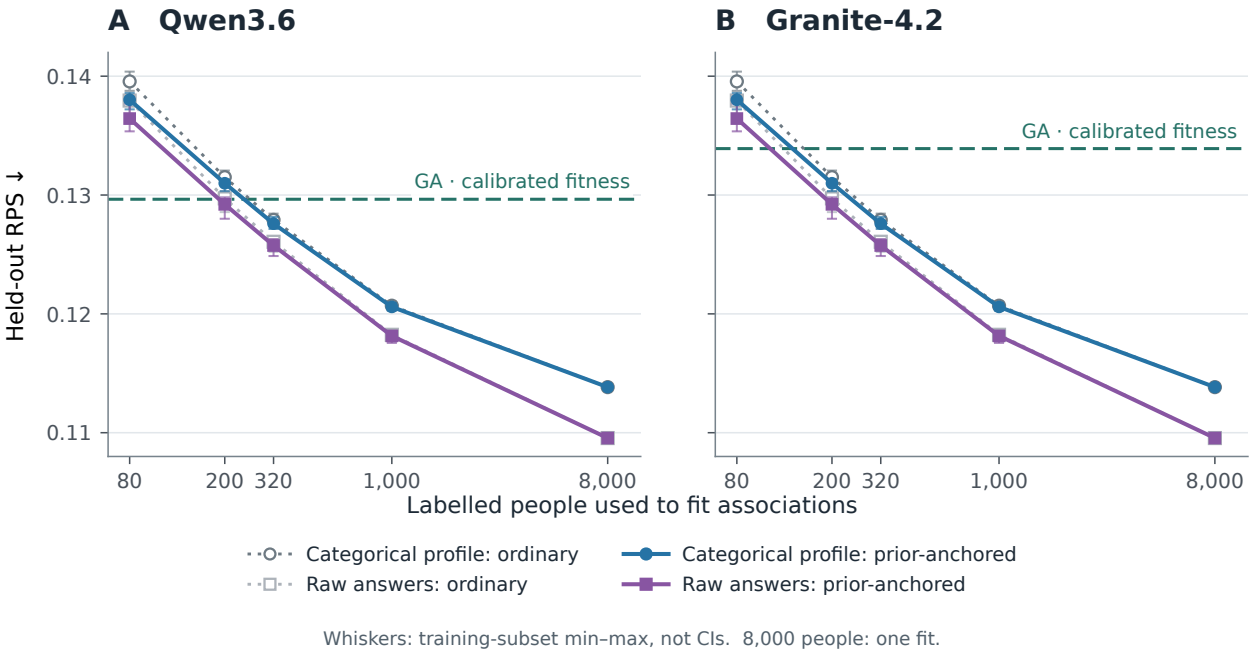


Figure S4. Ordinary and shared-prior ridge learning curves. Whiskers are ranges across ten overlapping fitted subsets, not confidence intervals. At 8,000 there is one fit. LLM lines average three frozen runs; statistical curves repeat across model panels for comparison.

18 Exploratory complementarity protocol

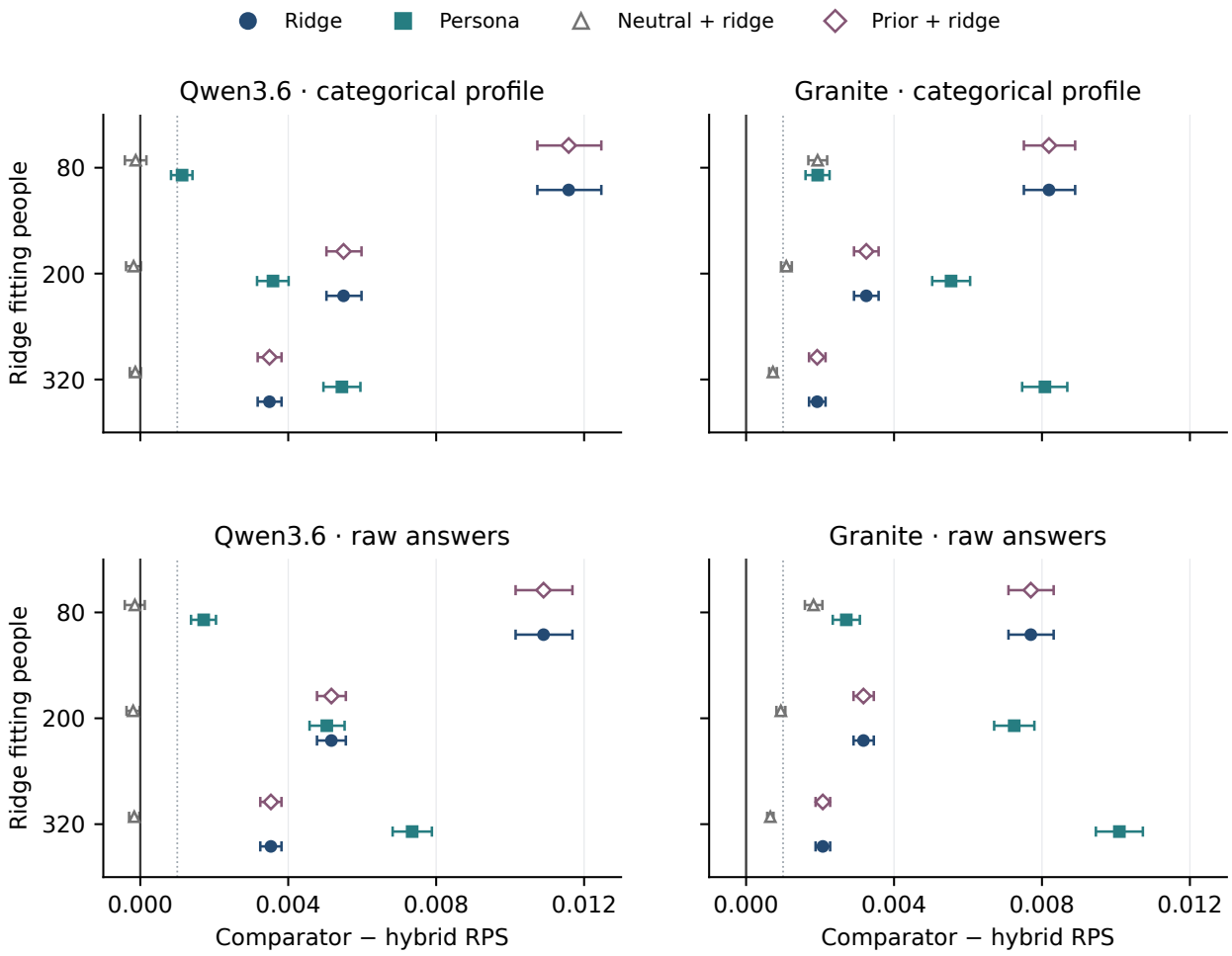
We apply the established linear forecast pool [1]:

$$H_{ij} = (1 - w)R_{ij} + wP_{ij}, \qquad w \in \{0, .05, \ldots, 1\}. \tag{2}$$

The convex combination is a valid CDF because both components are valid CDFs. Ridge uses categorical or raw-answer features and nested subsets of 80, 200 or 320 training people, defined by seeds 202610081–202610083. It selects its penalty on Select120. The first 60 ordered Calibrate people fit each persona's scale and mean CDF; the last 60 choose the mixture weight. Neither calibration nor mixture weights are refitted after selection. All three calibrated-GA runs and their own neutral counterparts are retained. No new LLM calls are made.

The empirical prior uses Train8000 and the original prompt search uses Search80/Select120. Each mixture requires the same last60 selection labels; neutral and population mixtures independently select their weights. Thus all supervision roles remain explicit. Train subsets overlap Train8000 and do not represent additional distinct people. RPS first averages 60 items per person, then averages losses over nine fixed fit/run combinations. This is not an ensemble of nine prediction vectors. A single 48-test Holm family covers all 12 conditions against four controls. Intervals and sign flips use 5,000 paired respondent draws with seed 20261008, conditional on these fixed fits. A useful screen result requires gain at least 0.001, a positive 95% interval and Holm $p < .05$. All comparisons and nulls are retained. Selection parameters were saved before explicit evaluation extraction; the full corpus and evaluation-bearing result containers had already been loaded. This is a procedural dataflow guard, not capability-level isolation; original full120-calibrated losses reproduce exactly in all 12 base/evolved checks.

19 Complete complementarity results



Positive favors hybrid; 95% respondent intervals conditional on fixed fits. Dotted line: 0.001 screen threshold.

Figure S5: All 48 contrasts: control minus optimized-persona mixture RPS. Positive favors the mixture. Bars are respondent-bootstrap 95% intervals conditional on the fixed nine fits; the dotted line is the 0.001 screening threshold. Color and marker identify the comparator. Full numerical means, intervals and within-family Holm values are retained in the associated table.

Mixtures beat both standalone components in every condition. Optimization itself passes the additional magnitude/statistical gate over the neutral mixture only on Granite for categorical 80/200 and raw 80. No Qwen condition passes that gate. The selected prior-mixture weights are all zero. Prior mixing is a different comparator from refitting the intercept around the population prior; the latter is tested separately below. The analysis is adaptive on exposed respondents and complements, rather than supersedes, earlier corpus-specific null evidence on persona combinations.

20 Adversarial prior-anchored comparison

After viewing the original screen, a separate protocol froze all 12 conditions and the original hybrid forecasts. The two new rivals are the prior-anchored ridge defined above, selecting its penalty on Select120, and a neutral-persona mixture using that ridge. Neutral calibration uses the first 60 Calibrate people and its independently selected mixture weight uses the last 60. All 18 anchored fits reproduce the corresponding earlier individual losses within 8.4×10^{-17}; original hybrid means are unchanged exactly. No new LLM calls or GPU inference are used. The procedural evaluation guard does not restrict the process's underlying file access.

The 24 contrasts form a separate Holm family with 5,000 paired respondent bootstrap/sign-flip draws, conditional on the same nine fixed combinations. The unchanged screening rule requires gain at least 0.001, positive marginal 95% interval and adjusted $p < .05$. All twelve hybrids beat anchored ridge by 0.00161–0.00984 RPS (all $p_{\mathrm{H}} = .0048$). Against neutral-plus-anchored ridge, Granite gains 0.00044–0.00164; only categorical 80 and raw 80 also clear the magnitude threshold. Qwen loses 0.00025–0.00038 in all six conditions; five differences have $p_{\mathrm{H}} < .05$ and categorical 80 has $p_{\mathrm{H}} = .0514$. Thus the stronger rival preserves the component-combination result while narrowing the case for optimized personas. This remains adaptive exposed-cohort evidence. Changing both the prompt and decoder makes the rival a practical comparison, not a causal ablation of optimization.

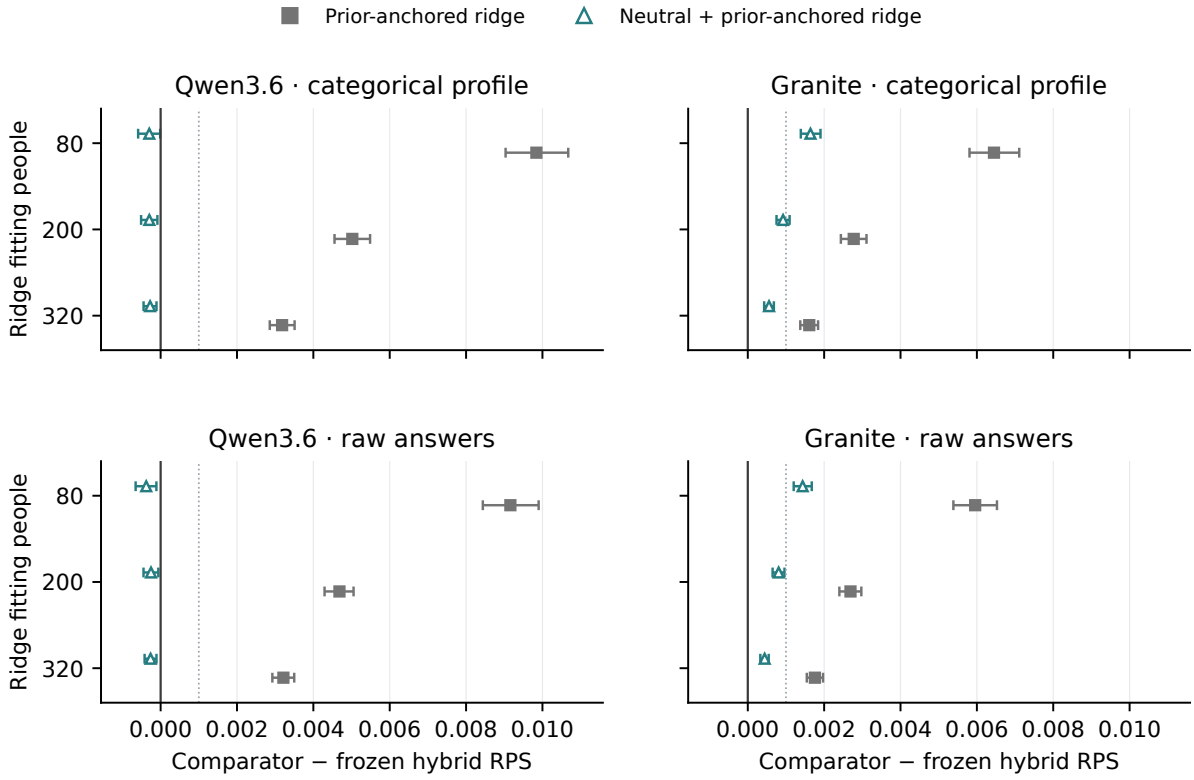


Figure S6: All 24 adversarial contrasts: comparator minus the unchanged original hybrid. Bars are conditional respondent-bootstrap 95% intervals; positive favors the hybrid. The dotted line is the 0.001 screening threshold. All conditions and unfavorable outcomes are retained; this family is separate from the original 48 contrasts.

21 Reproducibility specification

Common execution contract. Every cell used the same Qwen/Qwen3.8-27B mutator, a strict mutable JSON genome, temperature 0.7, at most 2,500 generated tokens per ordinary mutation, disabled thinking, and a fresh seeded mutator request. The serialized genome has exactly four top-level fields: `role_definition`, fixed-length arrays of five `trait_formulations` and 30 `facet_formulations`, and `critic_formulations`. The runtime genotype also carries fixed intensity modifiers copied from the neutral template; they are never mutated. Malformed proposals consumed no forecast budget, whereas any mutator transport failure invalidated the cell. Search was capped at $B = 80$ full optimization-panel equivalents. Since the panel has 80 respondents and 60 target items, the cap is $80 \times 80 \times 60 = 384{,}000$ respondent–item forecasts per cell; realized search usage ranges from 78.2 to 80.0 equivalents. The separate shortlist stage rescored at most five candidates on 120 respondents (7,200 forecasts per candidate; at most 36,000). It selected the lowest raw RPS with the genotype hash as a deterministic tie break. Thus the `generations`, `rounds`, and `iterations` fields below describe the planned schedule; the forecast ledger, rather than those fields, was the operative stopping rule.

Table S18: Exact initialization, update/history, and budget-relevant parameters for the eleven search methods. "Seed" is the neutral genotype specified below; all fitness comparisons minimize the cell's declared optimization objective.

Method	Initialization	Update and retained history	Fixed parameters
GA	Seed plus mutations to population 8.	Two elites; two independent tournament-3 parents. LLM crossover with probability .8 when parents differ; a crossed child is mutated with probability .3 and a copied parent is always mutated.	Population 8; elites 2; tournament 3; crossover .8; mutation .3; configured 10 generations.
Differential evolution	Same seed-plus-mutations population as GA.	For each member x, sample three distinct other members a, b, c; the LLM applies the $a - b$ difference to c and crosses the result with x; greedily replace x only on improvement.	Population 8; configured 10 generations. The recorded crossover-rate field is .8, but the deployed loop does not use it as a stochastic gate.
PromptBreeder	Seed, then seven mutations using successive members of a fixed eight-instruction mutation-prompt pool.	One elite; tournament-3 parent; each member carries its mutation instruction. With probability .25 the LLM rewrites that instruction (temperature .9, 200 tokens), then applies it to the parent genotype.	Population 8; instruction pool 8; hypermutation .25; configured 10 generations.
MAP-Elites style	Two islands, four initial evaluations per island; only the first is the unchanged seed, all others are seed mutations.	Uniformly sample an occupied archive cell; mutate using the last six lineage records (operation, score, prompt length, predicted SD); replace a cell only on lower score; migrate the island champion every 12 iterations.	2 islands; 4×4 bins with length edges $[0, 1500, 2500, 4000, 100000]$ and SD edges $[0, .8, 1, 1.2, 9]$; 4 initial points/island; history 6; configured 72 iterations.
Local GEPA	Evaluate seed on the full panel and initialize a respondent-wise Pareto pool.	Select a Pareto member in proportion to its number of respondent-wise wins; compare parent and reflected child on the same eight respondents; give the reflector the ten worst target-item errors; run a full evaluation only after a minibatch improvement. Retain the Pareto front and global best; spend remaining whole equivalents on best-candidate mutations.	Minibatch 8; at most 12 Pareto candidates; 10 feedback items; configured 80 evaluations.

Table S19: Search specifications, continued. The same executed-forecast budget applies.

Method	Initialization	Update and retained history	Fixed parameters
Official GEPA	Seed candidate and optimization respondents indexed 0–79.	GEPA v0.1.4 Pareto selection with per-example metric $-\mathrm{RPS}_{raw}$; reflection minibatches of 8; merge disabled; no metric cache. The external forecast ledger is authoritative and returns the last fully scored elite at exhaustion.	Reflection minibatch 8; package v0.1.4.
Native GigaEvo	Exact neutral JSON written as the initial program.	Pinned native guided pipeline; single island without distant parents; no memory; fitness $1 - \mathrm{RPS}$; external validator and forecast stopper.	Internal v2.20.1 at commit `f72ab08c`; maximum 160 mutants; concurrency 1; 8×8 behavior space over length $[0, 16000]$ and predicted SD $[0, 2]$.
OPRO	Evaluate seed.	Propose in blocks of 8 from up to the 12 best valid scored genotypes, printed from worse to better within that retained set.	8 proposals/round; history 12; configured 10 rounds.
ProTeGi	Evaluate seed and start a beam.	For every beam member, generate two textual-gradient edits using the ten worst target-item errors; evaluate one edit per gradient; retain the best four candidates.	Beam 4; 2 gradients per candidate; 1 edit/gradient; configured 20 rounds.
Random search	Evaluate seed.	Draw $B-1$ independent mutations of the seed; return the best full-panel candidate.	80 evaluations.
Best-of-B	Evaluate seed.	Draw $B-1$ independent seed mutations with the initial-population generator at temperature 1.0; return the best full-panel candidate.	80 evaluations; mutation temperature 1.0.

Neutral genotype and exact rendering. The neutral role is: "You are a simulated respondent. Answer each statement according to the supplied personality profile." For each of the five canonical traits $t \in \{\text{neuroticism, extraversion, openness, agreeableness, conscientiousness}\}$, its exact formulation is "Tendency toward t." For each canonical facet label f in the item key, the exact formulation is "Tendency toward lowercase(f)." The critic is: "Use only the supplied profile. Interpret the response scale consistently." Scores use boundaries $[0, 20, 40, 60, 80, 100]$ with labels *very little*, *slightly*, *moderately*, *quite strongly*, and *very strongly*; operational intervals are $[0, 20)$, $[20, 40)$, $[40, 60)$, $[60, 80)$, and $[80, 100]$ after endpoint clipping.

For a genotype (r, T, F, c), the exact system-prompt skeleton is `r\nYour profile:\n`, followed first by one line per trait and then one per facet: `- This trait (NAME) describes you MODIFIER: DESCRIPTION` and `- This facet (NAME) describes you MODIFIER: DESCRIPTION`. It ends with `\n\nInternal reflection guideline:\n` and c. A facet score is $100(\bar{x} - 1)/4$ from its two observed items; a trait score is the mean of its six facet scores.

Selected wording example. The frozen Qwen calibrated-GA seed-1 template changes the neutral "Tendency toward extraversion." to "The intensity with which you derive energy from social interaction, assertiveness, and positive emotional experiences." Its critic includes: "Avoid converging on a central tendency; instead, maximize the semantic variance between answers to accurately reflect the specific magnitude indicated, ensuring that a 'very little' response is distinctly different from a 'slightly' one in both tone and content, so that your agreement or disagreement scales linearly with the provided intensity." These are exact excerpts from one selected template, showing the intervention on profile interpretation. They neither characterize every optimized template nor establish a causal mechanism. The complete genotype accompanies the source notes.

Items, people, and partitions. Items are grouped by facet after sorting by item number. The first two items of each of 30 facets form the 60-item observed input; the remaining two form the disjoint 60-item prediction target. Eligible rows have all 120 item values in $\{1, 2, 3, 4, 5\}$, unique case IDs, and no match to the inventoried historical exposure set. Rows are sorted by case ID, permuted with seed 20261004, and sliced without overlap into train/optimization/selection/calibration/evaluation panels of 8,000/80/120/120/600 people (8,920 reserved rows). A subsequent inventory records 410,168 complete cases, of which 374,049 match the accumulated exclusion set and 36,119 remain eligible in that inventory. These are inventory counts, not a claim of new untouched test people; all-host exposure coverage is unproven. The local reproducibility bundle does not include the raw corpus, a verified dataset license, or validated demographic summaries. Sex, age, and country are loadable fields, but no demographic interpretation or distribution is asserted here.

Frozen exemplar and cryptographic closure. The saved Qwen calibrated-GA seed-1 selection has raw selection RPS 0.1692038488, five shortlisted candidates, and canonical selected-record hash `4131b9d1dc37cd4f`; its complete runtime genotype is transcribed in the accompanying source notes. A second located exemplar, Granite calibrated MAP-Elites seed 1, has selected-record hash `09d79eb1c951f4f8`. The canonical manifest hash is `9037e61283d70779`; panel hash `786de40844429e19`; frozen-record file hash `155b512315b52679`; final 84-cell recomputation hash `7a3009c03446254d`; model-lock file hash `342d9215e991e973`; item-key hash `2cd1483c7d3d539c`; and deployed source-closure hash `e333de246887ab0c`. Full 64-hex digests, model revision pins, source anchors, and the exact exemplar appear in `REPRO_SOURCE_NOTES.md`.

Statistical predictors and score translation. The 60 ordinal targets are encoded as 240 indicators $\mathbf{1}[y_j \le k]$ for $k = 1, \ldots, 4$. Raw inputs use 300 one-hot indicators; categorical inputs use 175. Ridge fits an intercept with the Cholesky solver, without scaling these one-hot

features. Select120 minimizes RPS over penalties $\{1, 10, 100, 1000\}$, with smaller penalties breaking ties. Continuous-profile features alone are standardized using training statistics. Predictions undergo the exact squared-error projection of each four-threshold vector onto $0 \le F_1 \le \cdots \le F_4 \le 1$. ExtraTrees uses 64 trees, minimum leaf size selected from $\{10, 30\}$, feature fraction .8, random seed 20261004, one worker, and otherwise scikit-learn defaults. Answer-equivalent curves use a nonincreasing isotonic fit and linear interpolation over the registered budget grid; out-of-grid values remain censored. Their bootstrap resamples respondents and the three fixed input orders, not training fits.

Pinned model revisions. The full model revision pins are:

Qwen predictor: `995ad96eacd98c81ed38be0c5b274b04031597b0`
Granite predictor: `9e668ce1c538387ef24d3644e9b0606647762636`
Qwen mutator: `1d4bf0f2ff6012fd82039f2fa52739d0dd7c60c0`.

The documented analysis environment uses Python 3.12.14, NumPy 2.3.5, scikit-learn 1.7.2, and serving vLLM 0.28.0. Full source notes and receipts are retained locally; no publicly executable anonymous artifact is claimed.

Table S20: All exploratory mixture conditions: mean RPS, lower is better. Each mean averages losses of the nine fixed fits. The preceding forest plot shows every paired contrast and its conditional interval.

Model	Features	Fit n	Hybrid	Ridge	Persona	Neutral mix	Prior mix
Qwen	categorical	80	0.12859	0.14017	0.12972	0.12846	0.14017
Qwen	categorical	200	0.12613	0.13163	0.12972	0.12595	0.13163
Qwen	categorical	320	0.12427	0.12776	0.12972	0.12413	0.12776
Qwen	raw	80	0.12800	0.13891	0.12972	0.12786	0.13891
Qwen	raw	200	0.12468	0.12984	0.12972	0.12448	0.12984
Qwen	raw	320	0.12237	0.12590	0.12972	0.12221	0.12590
Granite	categorical	80	0.13198	0.14017	0.13392	0.13392	0.14017
Granite	categorical	200	0.12838	0.13163	0.13392	0.12946	0.13163
Granite	categorical	320	0.12584	0.12776	0.13392	0.12656	0.12776
Granite	raw	80	0.13121	0.13891	0.13392	0.13303	0.13891
Granite	raw	200	0.12667	0.12984	0.13392	0.12761	0.12984
Granite	raw	320	0.12383	0.12590	0.13392	0.12448	0.12590

22 AI-assistance record

Codex assisted source inspection, experiment specification, coding and writing; retained review records identify GPT-5.6-Sol for draft and claim review. Retained Claude artifacts document generic figure-layout assistance; research labels and numerical values were bound and checked locally. Available records are inventoried in `AI_CONTRIBUTION_LEDGER_20261020.json` in the research evidence directory, including artifact hashes and coverage limits. This is a partial record: complete prompt/response retention and independently attested model identities are unavailable. Exact versions and prompts for older unrecorded interactions remain unknown.

23 Exploratory mixture uncertainty

24 Original-cohort objective effects

25 Detailed original-cohort budget and mixture diagnostics

25.1 Exploratory fitting-budget diagnostics

On the exposed cohort, shared-prior statistical controls have higher mean RPS than calibrated GA at 80 fitting people but lower mean RPS at 320; the supplement reports full learning curves and supervision counts.

Using the exact 80 search respondents, common-prior categorical ridge scores 0.13669. Calibrated GA is lower by 0.00705 on Qwen and 0.00278 on Granite. Against the stronger raw-answer control (0.13494), the Qwen difference is −0.00530 (crossed interval $[-0.00638, -0.00422]$, conditional Holm $p = .0096$); Granite's −0.00103 contrast does not satisfy the combined decision rule ($p = .0656$). Thus a restricted positive Qwen result survives the shared-prior check, whereas a cross-model claim against that control does not. The LLM still uses 120 additional calibration people and pretrained parameters; the ridge fitting count excludes its 120 selection people. These diagnostics neither establish equal-total-budget efficiency nor select an optimal budget on unexposed data.

Under the frozen ridge input curve, the primary Qwen persona corresponds to 10.97 answers (conditional 95% interval $[9.06, 12.92]$) and Granite to 7.51 ($[6.36, 8.84]$). These intervals propagate respondent and nested-order variation after averaging the three persona seeds; they do not propagate fresh optimizer-seed uncertainty. All 84 search curves reach 78.2–80 full-panel equivalents. Their checkpoints describe development efficiency, not a second procedure for selecting on evaluation respondents.

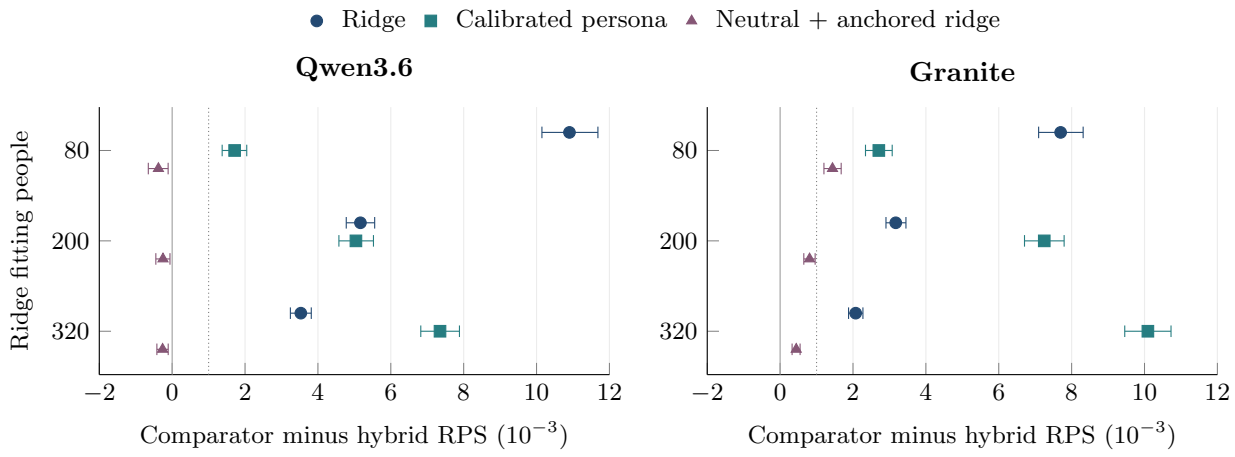


Figure S7: Exploratory paired gains on the exposed 600 respondents (raw-answer features). Positive values favor the optimized-persona mixture; bars are 95% respondent intervals conditional on nine fixed fit/run combinations. The vertical dotted line marks the 0.001 practical threshold. Component comparisons belong to the 48-test family; the stronger neutral rival to the separately frozen 24-test family. Intervals are pointwise. All LLMs also use the 8,000-person prior; small fitting counts are not total supervision budgets. The supplement includes categorical features and every declared comparison.

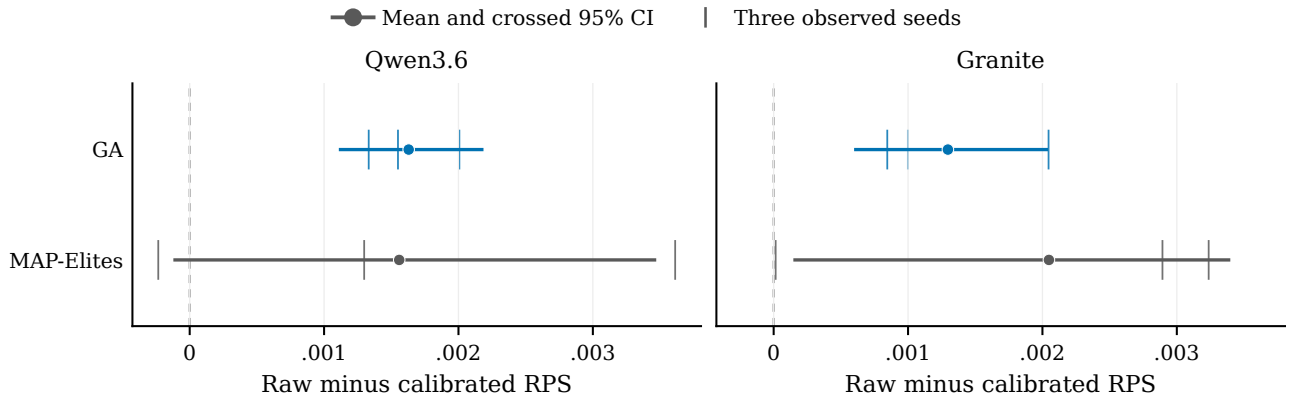


Figure S8: Direct objective effects, post-hoc analysis. Ticks show three paired seed effects; circles and bars show means and crossed 95% intervals. Positive values favor calibrated fitness. All observed GA seeds favor calibrated fitness, but three seeds do not establish generalization over optimizer randomness. Original comparisons remain in the main complete-comparison tables.

25.2 Exploratory complementarity

The selected persona–ridge mixture improves over both components in all 12 conditions under the screen rule. Gains over ridge range from 0.00192 to 0.01158 RPS, and gains over the calibrated persona from 0.00113 to 0.01009 (all conditional Holm $p = .0096$). The supplement shows paired gains and uncertainty for both representations and all 48 contrasts. No new model inference was needed: the experiment reuses saved forecasts, with calibration fitting and mixture selection separated.

The selected mixtures had lower RPS on this exposed cohort; the neutral control separates this observation from optimizer value. On Granite, the optimized mixture has lower mean RPS than the neutral mixture in all six conditions; three pass the 0.001 magnitude threshold as well as the statistical rule (categorical 80/200, raw 80). On Qwen, all six neutral mixtures have slightly lower means, with no significant difference after correction. Population-mixture selectors choose zero prior weight throughout, reproducing ridge. This linear-pool control is distinct from the earlier prior-anchored intercept refit. The observed advantage of combining predictors spans both tested models, while additional utility from persona search remains model- and budget-specific.

A separately frozen adversarial extension tests prior-anchored ridge and neutral-plus-anchored ridge in all 12 conditions (24-test Holm family). The original hybrid beats anchored ridge throughout, by 0.00161–0.00984 RPS (all $p_{\mathrm{H}} = .0048$). Against the stronger neutral mixture, only Granite categorical/raw 80 clear the magnitude/statistical rule. Qwen loses in all six conditions, five significantly. This practical rival changes both prompt and decoder; it is not a pure optimizer ablation or independent confirmation.

26 Detailed saved-forecast protocols

After inspecting the original evaluation, we fixed the following saved-forecast analyses before executing them. They reuse the exposed 600-person cohort and are *post hoc*, not independent confirmation. Within these diagnostics, no prompts are changed and no new LLM predictions are generated. The original 110-test analysis is retained.

Representation controls compare the exact 175 one-hot profile features visible in prompts with 35 continuous pre-discretization values and raw history. All use Train8000 and Select120; original losses reproduce within 10^{-10}. Separate Holm families cover 33 representation contrasts and nine objective/optimizer contrasts. Rank diagnostics retain the first complete evaluation per genotype in all 24 GA/MAP search pools.

Budget diagnostics fit categorical and raw-history ridge at 80, 200, 320, 1,000 and 8,000 people and exact Search80. Ten fixed nested subsets per smaller budget share Select120 penalty choice. A subsequent prior-anchored control replaces the subset target mean with the Train8000 prior. Both curves reproduce full-data losses. Each has a separate 48-test family; intervals condition on fits and resample respondents and observed LLM runs. Subset ranges are not confidence intervals. Full specifications and the adaptive exposure history are in the supplement.

Exploratory forecast combination. A final fixed screen tests complementarity using a linear pool of ordinal CDFs [1], $H = (1 - w)R + wP$, where R is ridge and P a calibrated-GA persona. Both models, both ridge representations and fitting sizes 80, 200 and 320 are included. Three fixed nested training subsets cross all three saved optimizer runs. Ridge penalties use Select. The first 60 Calibrate people fit the persona scale and center; the remaining 60 select $w \in \{0, .05, \ldots, 1\}$ by RPS. Parameters are fixed before evaluation. Controls are ridge, persona, neutral-persona plus ridge, and population-prior plus ridge; each mixture tunes its own weight on the same 60 people. A separate 48-test Holm family and 5,000 respondent-bootstrap draws condition on the nine fixed fit/run combinations. The screen requires mean gain at least 0.001, a positive gain interval and adjusted $p < .05$. All 12 conditions are reported. This adaptive analysis reuses the exposed cohort and the 8,000-person prior; low ridge fitting counts do not mean low total supervision.

27 Detailed representation contrasts

The additional categorical-profile ridge scores 0.11384, compared with 0.11558 for continuous-profile ridge and 0.10955 for raw-history ridge (Supplementary Figure S3). Its deficit to raw-history ridge is 0.00428 (paired respondent interval $[0.00378, 0.00478]$), while every LLM arm is 0.01581–0.02388 worse than categorical ridge; all 28 intervals are positive with Holm $p < .01$ in the 33-test family. Neutral personas also lose by 0.01625 on Qwen and 0.02273 on Granite. The profile retains information usable by a simple predictor even when persona search adds little. Ridge fits 8,000 person-level labelled mappings; the LLM pipeline uses those labels for item priors, while prompt optimization uses 80 search and 120 selection people. Thus person-level supervision budgets are not matched. This does not equate their learning procedures or give a causal compression/readout decomposition.

Relative to the empirical prior, categorical ridge retains $(0.15174 - 0.11384)/(0.15174 - 0.10955) = 89.8\%$ of raw-history ridge's RPS improvement. This is an observed prediction-improvement ratio, not a fraction of human information. Calibrated GA and MAP-Elites recover only 2.1% and 1.6% of their base-to-raw-ridge gaps on Qwen and 9.8% and 9.9% on Granite. These descriptive ratios have no additional significance claim.

References

[1] Tilmann Gneiting and Roopesh Ranjan. 2013. Combining Predictive Distributions. *Electronic Journal of Statistics* 7 (2013), 1747–1782. doi:10.1214/13-EJS823

[2] John A. Johnson. 2014. Measuring Thirty Facets of the Five Factor Model with a 120-Item Public Domain Inventory: Development of the IPIP-NEO-120. *Journal of Research in Personality* 51 (2014), 78–89. doi:10.1016/j.jrp.2014.05.003